Frontier



University of Hawai’i at Mãnoa

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

University of Hawai’i at Mãnoa

Organization location (we welcome applicants from anywhere in the world)

Honolulu, Hawai’i, USA

Name(s) of primary point(s) of contact for this application

David Ho, Lennart Gerke

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

University; school focused on ocean, earth, and environmental sciences

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Constraining nearshore air-sea gas exchange to enable reliable MRV of mCDR deployments

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Most marine carbon dioxide removal (mCDR) deployments will be conducted from the coast due to proximity to energy, infrastructure, and human resources. We have found that the relationship between wind speed and air-sea gas exchange differs significantly in nearshore environments from that in the open ocean. Hence, our project aims to determine whether nearshore air-sea gas exchange can be parameterized using a universally applicable relationship, similar to the one used in the open ocean. This would enable more robust and scalable monitoring, reporting, and verification (MRV) for marine carbon dioxide removal (mCDR) pathways such as ocean alkalinity enhancement (OAE) or direct ocean removal (DOR). This research directly advances the priority innovation area of improving the MRV toolkit for open-system pathways.

We will generate high-quality field observations of gas transfer velocities in a nearshore environment by conducting a dual tracer ($^3He/SF_6$) release experiment in Bedford Basin, an active marine carbon dioxide removal (mCDR) deployment site, along with physical measurements to determine the factors that control gas exchange in these settings. Accurate knowledge of air-sea gas exchange is fundamental to MRV because CDR occurs when CO_2 is transferred from the atmosphere into the ocean across the air-sea interface. Currently, all oceanographic models used to determine this flux employ open ocean wind speed parameterizations (e.g., Ho et al., 2006), which substantially overestimate gas transfer in coastal and nearshore environments by up to 65% (Dobashi et al., 2026; Gerke et al., in review). In these settings, wind fetch limitation, younger wave ages, and surfactants can substantially alter gas transfer velocities, introducing significant errors into carbon accounting for coastal mCDR deployments. Recent modeling work in Bedford Basin (Laurent et al., 2026) demonstrated that up to ~70% of CO_2 uptake following a release simulation can occur in the nearshore region, where an open ocean gas exchange parameterization will not be appropriate. Using open ocean parametrizations in these cases is likely to result in uptake estimates that are substantially overestimated.

Recent gas exchange studies that we have conducted with ^3He and SF_6 across diverse nearshore environments, such as Florida Bay (Dobashi and Ho., 2023), the Baltic Sea (Dobashi et al., 2026), an Icelandic fjord (Gerke et al., in review), and a coastal Salt Pond (Gerke et al., in prep.), consistently indicate that gas transfer velocities in nearshore regions are lower than the predictions from open ocean parameterizations (see Fig. 1). However, these observations remain sparse, site-specific, and there are currently insufficient data to establish a robust, universal parameterization for nearshore air-sea gas exchange. This lack of empirical constraint is a key bottleneck to accurate MRV and the scalable deployment of mCDR pathways.

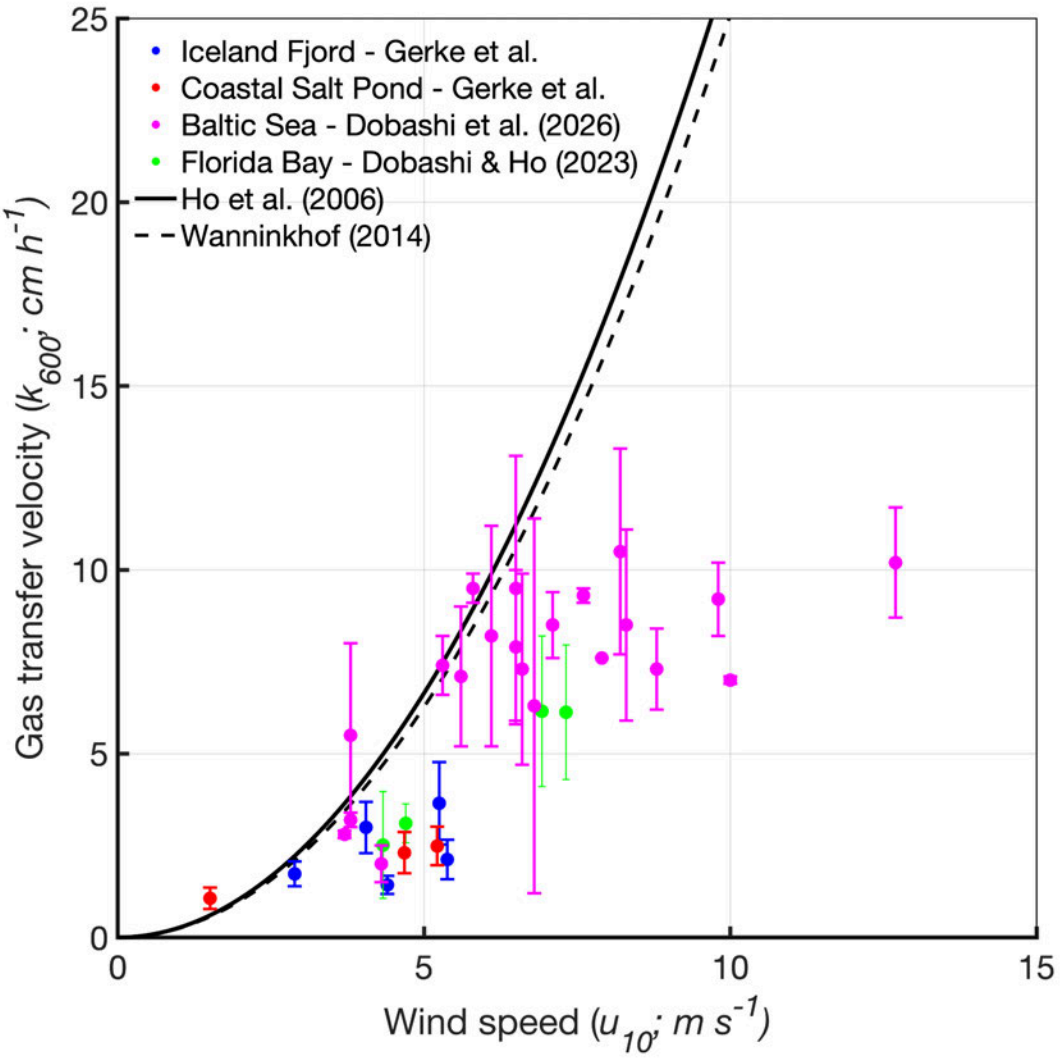


Figure 1: Observed gas transfer velocities (k_{600}) in coastal regions with correlating wind speeds at 10 m height (u_{10}). Parameterizations from open ocean experiments are shown as solid and dashed lines.

This project represents a critical next step toward identifying a universal parametrization for nearshore gas exchange by combining new, directly measured gas transfer velocities with observations of physical, chemical, and biological drivers, and integrating these observations with existing datasets across diverse coastal environments. This combined approach will result in one of two outcomes:

i. The development and validation of a parameterization that can be used to model gas exchange in diverse coastal deployment sites, while providing insight into which parameters should be monitored to reliably constrain the gas transfer velocity.
ii. The determination that site-specific observations are required for MRV applications, and to

identify the key observations needed.

With either outcome, this project will increase confidence in verified CO_2 removal and, by reducing uncertainty in air-sea gas exchange, make a critical step toward scalable and cost-effective mCDR MRV.

References:

Dobashi, R. and Ho, D.T., (2023), Air–sea gas exchange in a seagrass ecosystem – results from a $^3He/SF_6$ tracer release experiment, *Biogeosciences*, 20, 1075–1087, https://doi.org/10.5194/bg-20-1075-2023.

Dobashi, R., Ho, D.T., Marandino, C.A., and Schlosser, P. (2026), Air-sea gas exchange in the coastal Baltic Sea: Implications for marine carbon dioxide removal. *Journal of Geophysical Research: Oceans*, *131*, e2025JC023324. https://doi.org/10.1029/2025JC023324.

Gerke, L., Ho, D.T., Heede, U., and Koffman, T. (in review), Limits on marine carbon dioxide removal potential set by coastal air-sea gas exchange rates. *Comm Earth & Environ*.

Gerke, L. and Ho, D.T. (in prep.), Air-sea gas exchange in a coastal Salt Pond during an opportunistic limestone addition experiment.

Ho, D.T., Law, C. S., Smith, M.J., Schlosser, P., Harvey, M., and Hill, P. (2006), Measurements of air–sea gas exchange at high wind speeds in the Southern Ocean: Implications for global parameterizations, *Geophys. Res. Lett*., 33, L16611, https://doi.org/10.1029/2006GL026817.

Laurent, A., Wang, B., Atamanchuk, D., Rakshit, S., Azetsu-Scott, K., Algar, C., and Fennel, K. (2026), A high-resolution nested model to study the effects of alkalinity additions in Halifax Harbour, a mid-latitude coastal fjord, *Biogeosciences*, 23, 115–135, https://doi.org/10.5194/bg-23-115-2026.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

As outlined in the project summary, this project addresses a critical gap in MRV for mCDR pathways: the lack of a robust parameterization for nearshore air-sea gas exchange.

We will integrate direct field observations from the Bedford Basin with existing dual tracer datasets across coastal environments (e.g., Dobashi and Ho, 2023; Dobashi et al., 2026; Gerke et al., in review; Gerke et al., in prep.) to quantify gas exchange at this site and determine if governing processes can be parameterized reliably with a minimum set of variables to constrain a universal formulation across nearshore environments.

To achieve this, the project will address the following core research questions:

- ➔ What are the systematic differences between air-sea gas transfer velocities in nearshore environments and the open ocean parameterizations currently used in mCDR MRV frameworks?
- ➔ What physical, chemical, and biological processes control gas transfer in coastal environments, and which variables are essential for parameterizing nearshore gas exchange? We will assess the relative importance of wind speed, fetch, wave age, significant wave height, and surfactant presence in controlling gas transfer velocities.
- ➔ Over what spatial and temporal scales do site-specific gas exchange parameterizations remain valid before waters advect offshore and transition to open ocean regimes? This will require analysis of water mass residence times and transport.
- ➔ Can nearshore gas exchange observations be generalized into a universal parameterization, or are site-specific measurements required for robust MRV?

Our research site is the Bedford Basin in Nova Scotia, a semi-enclosed coastal basin where Planetary Technology is currently conducting an ocean alkalinity enhancement deployment. Dalhousie University has been studying the background biogeochemistry of this region for many years and is also documenting the effects of the Planetary alkalinity release. This site provides a unique and highly relevant setting to measure gas exchange directly within an active mCDR deployment. The core of our practical work will be a dual tracer ($^3He/SF_6$) release experiment in the Bedford Basin, alongside the necessary oceanographic and physical measurements.

The dual tracer method is the most robust and widely accepted field technique for quantifying air-sea gas transfer velocities over multi-day timescales. It involves the controlled release of two inert tracers (^3He and SF_6) with substantially different molecular diffusivities into the surface mixed layer at a known ratio. Because first-order physical advection and dispersion affect both tracers equally, changes in their concentration ratio over time can be attributed solely to gas exchange across the air-sea interface. This allows direct determination of gas transfer velocities independent of uncertainties associated with advection and dilution.

Following the tracer release, we will track the tracer patch through a combination of continuous and discrete measurements. Surface water SF_6 concentration will be mapped in near real time using a well-established underway sampling system (Ho et al., 2002), which provides high-resolution spatial coverage of tracer dispersion. Discrete water samples collected throughout the water column will be analyzed for both ^3He and SF_6 concentrations at regular time intervals (approx. every 12 hours), enabling calculation of gas transfer velocities from temporal changes in the tracer ratio.

These tracer observations will be complemented by comprehensive environmental measurements, including wind speed and direction, significant wave height, surface surfactant indicators, and conductivity, temperature, and depth (CTD) profiles to characterize the mixed layer depth. Simultaneous measurements of turbulent microstructure shear and temperature will be undertaken to distinguish the actively mixing surface layer and to quantify the near-surface turbulent kinetic energy dissipation rates that are expected to correlate with the gas transfer velocity. Together, these data will allow us to link observed gas transfer velocities to their controlling processes and to evaluate which variables are necessary and sufficient for parameterizing gas exchange in nearshore environments.

In addition to quantifying gas transfer, the tracer experiment will provide constraints on plume evolution and, thereby, on surface water residence times within the basin. By tracking the spatial distribution of SF_6, we will determine how long alkalinity-enhanced waters remain within the Bedford Basin and Halifax Harbor before advecting offshore into the Atlantic Ocean. This information is critical for defining the spatial and temporal domains over which nearshore gas exchange parameterizations

need to be applied, and when a transition to open ocean parameterizations is appropriate. The use of SF_6 as a tracer is essential for this application, as short-lived fluorescent tracers, such as Rhodamine WT, rapidly dilute below detection limits within hours and cannot resolve multi-day transport or basin-scale residence times.

Bedford Basin is particularly well-suited for this study because it captures key characteristics of many prospective coastal mCDR deployment sites. It is a semi-enclosed basin with restricted exchange to the open ocean, enabling multi-day tracking while still allowing eventual export to offshore waters. Its proximity to the Atlantic Ocean provides an ideal gradient from nearshore to open ocean conditions, making it well suited for studying the transition between these regimes. Due to the active mCDR deployment, the measurements collected in this project are directly relevant to current MRV challenges, as CO_2 uptake is estimated using models that employ open ocean parameterizations.

This project differs from prior coastal gas exchange studies in several key aspects. First, it is conducted within an active mCDR deployment setting. Second, the project explicitly targets the development of a universal parameterization that moves beyond site-specific characterizations by incorporating multiple environmental factors and analyzing the controlling processes. And third, the combined measurement of gas transfer velocities and surface water residence time provides a more complete framework for MRV at this site.

References:

Dobashi, R. and Ho, D.T. (2023), Air–sea gas exchange in a seagrass ecosystem – results from a ^3He / SF_6 tracer release experiment, *Biogeosciences*, 20, 1075–1087, https://doi.org/10.5194/bg-20-1075-2023.

Dobashi, R., Ho, D.T., Marandino, C.A., and Schlosser, P. (2026), Air-sea gas exchange in the coastal Baltic Sea: Implications for marine carbon dioxide removal. *Journal of Geophysical Research: Oceans*, *131*, e2025JC023324. https://doi.org/10.1029/2025JC023324.

Gerke, L.; Ho, D.T., Heede, U., Koffman, T. (in review), Limits on marine carbon dioxide removal potential set by coastal air-sea gas exchange rates. *Comm Earth & Environ*.

Gerke, L. and Ho, D.T. (in prep.), Air-sea gas exchange in a coastal Salt Pond during an opportunistic limestone addition experiment.

Ho, D. T., Schlosser, P., and Caplow, T. (2002), Determination of longitudinal dispersion coefficient and net advection in the tidal Hudson River with a large-scale, high resolution SF_6 tracer release experiment. *Environmental Science & Technology, 36*(15), 3234-3241, https://doi.org/10.1021/es015814+.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

The primary scientific and technical risk to the project's success is the potential loss or insufficient resolution of the injected tracer patch due to unexpected advection, enhanced vertical mixing, or rapid dilution. This has never happened in three decades of performing these experiments. Also, this risk is mitigated by the semi-enclosed nature of Bedford Basin, which limits large-scale dispersion,

and by the experimental design, which includes the option of a second tracer injection if necessary. In addition, to further mitigate this risk, prior hydrodynamic data within the area will be observed, and the experiment will be scheduled for favorable meteorological and tidal conditions. During the experiment, continuous surface SF_6 observations will enable adaptive ship-based tracking of the tracer plume.

A second technical risk is potential instrumentation failure that could affect tracer measurements. This will be mitigated through the use of field-proven instrumentation, pre-cruise calibration and testing, and the availability of onboard spare instruments and components. The project team has extensive experience conducting dual tracer release experiments and operating the required analytical systems, further reducing this risk.

A key scientific and MRV-related risk is that observed relationships between gas transfer velocities and environmental drivers may remain site-specific and not directly transferable to other coastal settings. This risk is inherent to the objective of developing a universal parameterization. However, the project explicitly focuses on identifying the physical, chemical, and biological controls on nearshore gas exchange, enabling process-based interpretation rather than purely empirical relationships to wind speed. Results will also be evaluated in the context of existing coastal datasets to assess broader applicability.

Other project execution risks include weather-related disruptions during the field trial. These will be mitigated by maintaining scheduling flexibility to adapt to environmental conditions and conducting extensive background analysis of seasonal climatologies.

Overall, while uncertainties are unavoidable in field-based experiments, the risks are low, and based on the team's extensive prior experience, are well-manageable.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Gas transfer velocities and universal parameterization presented in peer-reviewed publications, along with a validated dataset
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	All deliverables will directly address the uncertainty in quantifying and verifying atmospheric CO_2 uptake by mCDR deployments and improve site-specific MRV and models.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier*	Until Q3 2026, we will focus on project planning, experimental design, and instrument preparation.

generally prefers projects with clear readouts within ~18 months.)	During Q4 2026, we will execute the dual tracer release experiment. By Q1 2028, we will have all samples measured, quality controlled, and analyzed. By Q2 2028, we will provide a final report, publish the validated dataset, and submit manuscripts.
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$762,455